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Arthur Cayley

325. Note on a Theorem Relating to a Triangle, Line, and Conic *(continued)*

[Continuation from p. 100.]

and the equation of the line joining this point with the vertex $(y=0,\ z=0)$ is $(a\mu - h\lambda)\,y = (g\lambda - a\nu)\,z$. The equations of the three joining lines therefore are

$$\begin{aligned}(a\mu - h\lambda)\,y &= (g\lambda - a\nu)\,z,\\ (b\nu - f\mu)\,z &= (h\mu - b\lambda)\,x,\\ (c\lambda - g\nu)\,x &= (f\nu - c\mu)\,y,\end{aligned}$$

lines which will meet in a point if

$$(a\mu - h\lambda)(b\nu - f\mu)(c\lambda - g\nu) - (g\lambda - a\nu)(h\mu - b\lambda)(f\nu - c\mu) = 0,$$

or, multiplying out and putting as usual

$$\begin{aligned}K &= abc - af^2 - bg^2 - ch^2 + 2fgh,\\ \mathfrak{A} &= bc - f^2, \quad \&\text{c.},\end{aligned}$$

if

$$2(abc - fgh)\,\lambda\mu\nu + a\mathfrak{A}\,\mu\nu^2 + a\mathfrak{A}\,\mu^2\nu + b\mathfrak{B}\,\nu\lambda^2 + b\mathfrak{B}\,\nu^2\lambda + c\mathfrak{C}\,\lambda\mu^2 + c\mathfrak{C}\,\lambda^2\mu = 0,$$

that is, the line must touch a curve of the third class.

If this equation break up into factors, the form must be

$$(\alpha\lambda + \beta\mu + \gamma\nu)(A\mu\nu + B\nu\lambda + C\lambda\mu) = 0;$$

that is, we must have

$$\begin{aligned}A\alpha + B\beta + C\gamma &= 2(abc - fgh),\\ B\alpha = b\mathfrak{B}, &\quad C\alpha = c\mathfrak{C},\\ C\beta = c\mathfrak{C}, &\quad A\beta = a\mathfrak{A},\\ A\gamma = a\mathfrak{A}, &\quad B\gamma = b\mathfrak{B};\end{aligned}$$

and the last six equations give without difficulty

$$A = \frac{ka}{\mathfrak{A}}, \quad B = \frac{kb}{\mathfrak{B}}, \quad C = \frac{kc}{\mathfrak{C}}, \quad \alpha = \frac{1}{k}\mathfrak{B}\mathfrak{C}, \quad \beta = \frac{1}{k}\mathfrak{C}\mathfrak{A}, \quad \gamma = \frac{1}{k}\mathfrak{A}\mathfrak{B},$$

where k is arbitrary; the first equation then gives

$$\frac{a\mathfrak{B}\mathfrak{C}}{\mathfrak{A}} + \frac{b\mathfrak{C}\mathfrak{A}}{\mathfrak{B}} + \frac{c\mathfrak{A}\mathfrak{B}}{\mathfrak{C}} = 2(abc - fgh);$$

or, reducing by the equations $\mathfrak{BC} = a\mathfrak{A} + gh$, &c., this is

$$2\mathfrak{A}a + \mathfrak{B}b + \mathfrak{C}c - 2abc + 2fgh + (1 + 1 + 1)\frac{K}{1} = 0;$$

which, substituting for $\mathfrak{A}$, $\mathfrak{B}$, $\mathfrak{C}$ their values, becomes

$$K\left(1 + \frac{a^2}{\mathfrak{A}} + \frac{b^2}{\mathfrak{B}} + \frac{c^2}{\mathfrak{C}}\right) = 0.$$

Hence if $K = 0$, that is, if the conic break up into a pair of lines, or if

$$1 + \frac{a^2}{\mathfrak{A}} + \frac{b^2}{\mathfrak{B}} + \frac{c^2}{\mathfrak{C}} = 0,$$

in either case the equation of the curve of the third class becomes

$$\left(\frac{\lambda}{\mathfrak{A}} + \frac{\mu}{\mathfrak{B}} + \frac{\nu}{\mathfrak{C}}\right)\left(\frac{a}{\mathfrak{A}}\mu\nu + \frac{b}{\mathfrak{B}}\nu\lambda + \frac{c}{\mathfrak{C}}\lambda\mu\right) = 0;$$

that is, the curve breaks up into a point, and a conic inscribed in the triangle.

In the case where the conic breaks up into a pair of lines, then we have

$$(a,\ b,\ c,\ f,\ g,\ h \,|\, x,\ y,\ z)^2 = 2(px + qy + rz)(p'x + q'y + r'z),$$

and thence

$$(\mathfrak{A},\ \mathfrak{B},\ \mathfrak{C},\ \mathfrak{F},\ \mathfrak{G},\ \mathfrak{H} \,|\, x,\ y,\ z)^2 = -\{(qr' - q'r)x + (rp' - r'p)y + (pq' - p'q)z\}^2;$$

so that the equation in (λ, μ, ν) is

$$\{(qr'-q'r)\lambda+(rp'-r'p)\mu+(pq'-p'q)\nu\}\{pp'(qr'-q'r)\mu\nu+qq'(rp'-r'p)\nu\lambda+rr'(pq'-p'q)\lambda\mu\} = 0;$$

where the point represented by the equation

$$(qr' - q'r)\lambda + (rp' - r'p)\mu + (pq' - p'q)\nu = 0$$

is, of course, the intersection of the two lines.

326. Theorems Relating to the Canonic Roots of a Binary Quantic of an Odd Order.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 206–208.]

I CALL to mind Professor Sylvester's theory of the canonical form of a binary quantic of an odd order; viz., the quantic of the order $2n+1$ may be expressed as a sum of a number $n+1$ of $(2n+1)$th powers, the roots of which, or say the *canonic roots* of the quantic, are to constant multipliers *près* the factors of a certain covariant derivative of the order $(n+1)$, called the *Canonizant.* If, to fix the ideas, we take a quintic function, then we may write

$$(a,\, b,\, c,\, d,\, e,\, f \,|\, x,\, y)^5 = A\,(lx+my)^5 + A'\,(l'x+m'y)^5 + A''\,(l''x+m''y)^5$$

(it would be allowable to put the coefficients A each equal to unity; but there is a convenience in retaining them, and in considering that a canonic root $lx+my$ is only given as regards the ratio $l:m$, the coefficients l, m remaining indeterminate); and then the canonic roots $(lx+my)$, &c. are the factors of the Canonizant

$$\begin{vmatrix} y^3, & -y^2x, & yx^2, & -x^3 \\ a, & b, & c, & d \\ b, & c, & d, & e \\ c, & d, & e, & f \end{vmatrix}.$$

It is to be observed that this reduction of the quantic to its canonical form, i.e. to a sum of a number $n+1$ of $(2n+1)$th powers, is a unique one, and that the quantic cannot be in any other manner a sum of a number $n+1$ of $(2n+1)$th powers.

Professor Sylvester communicated to me, under a slightly less general form, and has permitted me to publish the following theorems:

1. If the second emanant $(X\partial_x + Y\partial_y)^2 U$ has in common with the quantic U a single canonic root, then all the canonic roots of the emanant are canonic roots of the quantic; and, moreover, if the remaining canonic root of the quantic be $rx + sy$, then (X, Y), the facients of emanation, are $= (s, -r)$, or, what is the same thing, they are given by the equation

$$\text{canont. } U \ (X, Y \text{ in place of } x, y) = 0.$$

In fact, considering, as before, the quintic $U = (a, b, c, d, e, f \mid x, y)^5$, we have

$$U = A(lx + my)^5 + A'(l'x + m'y)^5 + A''(l''x + m''y)^5,$$

and thence

$$(X\partial_x + Y\partial_y)^2 U = B(lx + my)^3 + B'(l'x + m'y)^3 + B''(l''x + m''y)^3,$$

if for shortness

$$B = 6 \cdot 5\,(lX + mY)^2 A, \ \text{\&c.}$$

Suppose $(X\partial_x + Y\partial_y)^2 U$ has in common with U the canonic root $lx + my$, then

$$(X\partial_x + Y\partial_y)^2 U = C(lx + my)^3 + C'(px + qy)^3,$$

and thence

$$B'(l'x + m'y)^3 + B''(l''x + m''y)^3 = (C - B)(lx + my)^3 + C'(px + qy)^3,$$

which must be an identity; for otherwise we should have the same cubic function expressed in two different canonical forms. And we may write

$$B' = C', \quad l'x + m'y = px + qy, \quad B'' = 0, \quad C = B,$$

and then we have

$$(X\partial_x + Y\partial_y)^2 U = B(lx + my)^3 + B'(l'x + m'y)^3;$$

so that all the canonic roots of the emanant are canonic roots of the quantic. Moreover, the condition $B'' = 0$ gives $l''X + m''Y = 0$, that is, $X : Y = m'' : -l''$, or writing $rx + sy$ instead of $l''x + m''y$, $X : Y = s : -r$; and the system is

$$\begin{aligned} U &= A(lx + my)^5 + A'(l'x + m'y)^5 + A''(rx + sy)^5, \\ (s\partial_x - r\partial_y)^2 U &= B(lx + my)^3 + B'(l'x + m'y)^3, \end{aligned}$$

which proves the theorem.

2. The two functions, canont. U, canont. $(X\partial_x + Y\partial_y)^2 U$, have for their resultant $\{\text{canont. } U\,(X, Y \text{ in place of } x, y)\}^{2n}$, if $2n + 1$ be the order of U.

In fact, in order that the equations

$$\text{canont. } U = 0, \quad \text{canont. } (X\partial_x + Y\partial_y)^2 U = 0,$$

may coexist, their resultant must vanish; and conversely, when the resultant vanishes, the equations will have a common root. Now if the equation canont. $(X\partial_x + Y\partial_y)^2U = 0$ has a common root with the equation canont. $U = 0$, all its roots are roots of canont. $U = 0$; and, moreover, if $rx + sy = 0$ be the remaining root of canont. $U = 0$, then $X : Y = s : -r$, that is, we have

$$\text{canont. } U\,(X,\, Y \text{ in place of } x,\, y) = 0;$$

or the resultant in question can only vanish if the last-mentioned equation is satisfied. It follows that the resultant must be a power of the *nilfactum* of the equation; and observing that canont. U is of the form $(a, \ldots)^{n+1}(x,\, y)^{n+1}$, i.e. that it is of the degree $n + 1$ as well in regard to the coefficients as in regard to the variables $(x,\, y)$, it is easy to see that the resultant is of the degree $2n(n + 1)$ as well in regard to the coefficients as in regard to $(X,\, Y)$; that is, we have $2n$ as the index of the power in question.

3. In particular, if $Y = 0$, the theorem is that the resultant of the functions canont. U, canont. $\partial_x^2 U$ is equal to the $2n$th power of the first coefficient of canont. U.

Thus for $n = 1$, that is, for the cubic function $(a,\, b,\, c,\, d \,|\, x,\, y)^3$, we have

$$\text{canont. } U = \begin{vmatrix} y^2, & -xy, & x^2 \\ a, & b, & c \\ b, & c, & d \end{vmatrix} = (ac - b^2,\ ad - bc,\ bd - c^2 \,|\, x,\, y)^2,$$

$$\text{canont. } \partial_x^2 U = \begin{vmatrix} y, & -x \\ a, & b \end{vmatrix} = ax + by;$$

and the resultant of the two functions is

$$= (ac - b^2,\ y,\ ad - bc,\ bd - c^2 \,|\, b,\, -a)^2$$

$$= -(ac - b^2) \cdot a^2,$$

which verifies the theorem.

The theorems were, in fact, given to me in relation to the quantic U and the second differential coefficient $\partial_x^2 U$; but the introduction instead thereof of the second emanant $(X\partial_x + Y\partial_y)^2U$ presented no difficulty.

2, Stone Buildings, W.C., February 16, 1863.

327. On the Stereographic Projection of the Spherical Conic.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 350–353.]

In order to the tolerable delineation of some figures relating to spherical geometry, I had occasion to consider the stereographic projection of the spherical conic. To fix the ideas, imagine a sphere having its centre in the plane of the paper, and through the centre three rectangular axes, that of x horizontal and that of y vertical, in the plane of the paper, and the axis of z perpendicular to and in front of the plane of the paper. The radius of the sphere is taken equal to unity (so that its intersection by the plane of the paper is the circle radius unity), and the points X, Y, and Z are taken to denote the points where the axes, drawn in the positive direction, meet the surface of the sphere; and the opposite points are called X', Y', and Z'. The eye is supposed to be at Z, and the projection to be made on the plane of the paper. This being so, and supposing that the axes of coordinates are the principal axes of the spherical conic, the *axis of x being the interior axis*, and taking ξ, η, ζ as the coordinates of a point on the spherical conic, its equations are

$$\xi^2+\eta^2+\zeta^2=1, \qquad -\xi^2+\eta^2\cot^2\beta+\frac{\zeta^2}{c^2}=0;$$

where it may be remarked that $\tan\beta$, c are the semiaxes of the plane conic which is the gnomonic projection (i.e. the projection by lines through the centre of the sphere) of the spherical conic on the tangent plane at X or X'.

Taking, for a moment, x, y, z as the coordinates of a point on the projecting line (that is, the line through the eye to a point (ξ,η,ζ) on the spherical conic), the equation of this line is

$$\frac{x}{\xi}=\frac{y}{\eta}=\frac{z-1}{\zeta-1};$$

and thence, putting $z = 0$, x, y will be the coordinates of a point of the projection, and we have

$$\frac{x}{\xi} = \frac{y}{\eta} = \frac{1}{1-\zeta};$$

or, what is the same thing,

$$\xi = x(1-\zeta), \quad \eta = y(1-\zeta);$$

the equations of the spherical conic may be written

$$\begin{aligned} 1-\zeta^2 &= \xi^2+\eta^2, \\ \zeta^2 &= c^2(\xi^2-\eta^2\cot^2\beta); \end{aligned}$$

and by eliminating ξ, η, ζ from the four equations, we obtain the equation of the conic.

Substituting for ξ and η their values, we find

$$1+\zeta = (x^2+y^2)(1-\zeta),$$

$$\zeta^2 = c^2(x^2-y^2\cot^2\beta)(1-\zeta)^2;$$

or, observing that the first equation gives

$$\zeta = \frac{x^2+y^2-1}{x^2+y^2+1},$$

and that thence

$$1-\zeta = \frac{2}{x^2+y^2+1}, \qquad \frac{\zeta}{1-\zeta} = \tfrac{1}{2}(x^2+y^2-1),$$

the equation is

$$(x^2+y^2-1)^2 = 4c^2(x^2-y^2\cot^2\beta).$$

It is now very easy to trace the curve. We see first that the curve is symmetrical with respect to the axes, and that it meets the axis of y in four imaginary points, but the axis of x in four real points, the coordinates whereof are

$$x = \pm(\sqrt{1+c^2}\pm c),$$

so that the two points on the same side of the centre are the images one of the other in regard to the circle radius unity. Moreover the curve touches the lines

$$y = \pm x\tan\beta$$

at their intersections with the circle. By developing in regard to y, the equation becomes

$$y^4 + 2(x^2-1+2c^2\cot^2\beta)y^2 + (x^2-1)^2 - 4c^2x^2 = 0;$$

and putting

$$x = \pm(\sqrt{1+c^2}\pm c),$$

the last term vanishes, and the equation gives $y^2 = 0$, or

$$\begin{aligned} y^2 &= 2(1 - x^2 - 2c^2 \cot^2 \beta) \\ &= 4(-c^2 \mp c\sqrt{1+c^2} - c^2 \cot^2 \beta) \\ &= 4c(-c \csc^2 \beta \mp \sqrt{1+c^2}), \end{aligned}$$

the upper sign corresponding to the exterior values

$$\pm x = \sqrt{1+c^2} + c,$$

and the lower sign to the interior values

$$\pm x = \sqrt{1+c^2} - c.$$

In the former case the values of y are imaginary; in the latter case they are real if

$$\sqrt{1+c^2} > c \csc^2 \beta,$$

or, what is the same thing, if

$$\sin^2 \beta > \frac{c}{\sqrt{1+c^2}};$$

that is, if (for a given value of c) β is sufficiently great, but otherwise they are imaginary.

If, as in the annexed figures, $c = \frac{5}{12}$ (and therefore $\sqrt{1+c^2} = \frac{13}{12}$, $\sqrt{1+c^2} + c = \frac{3}{2}$, $\sqrt{1+c^2} - c = \frac{2}{3}$), then for the limiting value of β we have

$$\sin^2 \beta = \tfrac{5}{13} = 0.3846, \quad \sin \beta = 0.62, \text{ or } \beta = 38^\circ \text{ nearly.}$$

[Fig. 1 and Fig. 2: figures of the curve for two values of β are omitted in this reset.]

In the first figure β is less, in the second figure greater than this value: the form for the limiting value is obvious from a comparison of the two figures.

I take the opportunity to mention the following theorem, which is perhaps known, but I have not met with it anywhere; viz. any three circles, each two of which meet, may be considered as the stereographic projections of three great circles of the sphere. In fact suppose, as above, that the projection is made on the plane of a great circle, and calling this the principal circle, the projection of any other great circle meets the principal circle at the extremities of a diameter of the principal circle. It follows that the theorem will be true, if, given any three circles each two of which meet, a circle can be drawn meeting the given circles, each of them at the extremities of a diameter of the circle so to be drawn. It is easy to see that the required circle has for its centre the radical centre (point of intersection of the radical axes) of the given circles, and that the radius is the 'Inner Potency' of the point in question in regard to each of the three given circles. In particular the three circles having for centres the vertices of an equilateral triangle, and the side for radius, may be considered as the stereographic projections of three great circles of a sphere. This is a very ready mode of delineation of a spherical figure depending on three great circles of the sphere.

2, Stone Buildings, W.C., March 21, 1863.

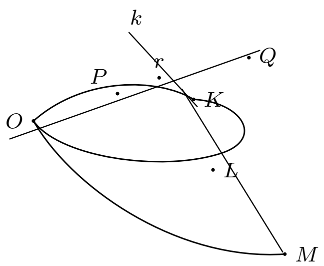
328. On the Delineation of a Cubic Scroll.

[From the *Philosophical Magazine*, vol. xxv. (1863), pp. 528–530.]

Imagine a cubic scroll (skew surface of the third order) generated by lines each of which meets two given directrix lines. One of these is a nodal (double) line on the surface, and I call it the nodal directrix; the other is a single line on the surface, and I call it the single directrix. The section by any plane is a cubic passing through the points in which the plane meets the directrix lines; i.e. the point on the nodal directrix is a node (double point) of the curve, the point on the single directrix a single point on the curve; the two directrix lines, and the cubic curve, the section by any plane, determine the scroll. Consider the sections by a series of parallel planes. Let one of these planes be called the basic plane, and the section by this plane the basic section or basic cubic; and imagine any other section projected on the basic plane by lines parallel to the nodal directrix: such section may be spoken of simply as 'the section,' and its projection as 'the cubic.' The cubic has at the node of the basic cubic a node, and besides has with the basic cubic four points in common. The two curves have, moreover, in common the three points at infinity (or, in other words, their asymptotes are parallel); in fact the points at infinity of either curve are the points in which the line at infinity, the intersection of the basic plane and the plane of the section, meets the scroll; and these points are therefore the same for each of the two curves. The remaining two points of intersection of the cubic with the basic cubic are also fixed points on the basic cubic, i.e. they are the points of intersection of the basic plane by the two generating lines parallel to the nodal directrix. Hence the cubic meets the basic cubic in nine fixed points, viz. the node counting as four points, the three points at infinity, and the two points the feet of the generators parallel to the nodal directrix. It follows that if $U = 0$ is the equation of the basic cubic, $V = 0$ the equation of some other cubic meeting the basic cubic in the nine points in question, then the equation of 'the cubic' is $U + \lambda V = 0$, λ being a parameter the value of which varies according to the position (in the series of parallel planes) of the plane of the section. Suppose that the basic cubic $U = 0$ is given, and suppose for a moment that the cubic $V = 0$ is also given, these two cubics having the above-mentioned relations, viz. they have a common node and parallel asymptotes: the cubic $U + \lambda V = 0$ might be constructed by drawing through the node (say O) a radius vector meeting the cubics in P, P' respectively, and taking on this radius vector a point Q such that $PQ = \dfrac{\lambda}{1+\lambda} PP'$, or, what is the same thing,

$$OQ = \frac{OP + \lambda OP'}{1+\lambda};$$

the locus of the point Q will then be the cubic $U + \lambda V = 0$. And we may even suppose the cubic $V = 0$ to break up into a line and a conic (hyperbola), and then (disregarding the line) use the hyperbola in the construction. In fact, if the hyperbola is determined by the following five conditions, viz. to pass through the node and through the feet of the two generators parallel to the nodal directrix, and to have its asymptotes parallel to two of the asymptotes of the basic cubic, and if the line be taken to be a line through the node parallel to the third asymptote of the basic cubic; then the hyperbola and line form together a cubic curve meeting the basic cubic in the nine points, and therefore satisfying the conditions assumed in regard to the cubic $V = 0$. And it is to be noticed that as in general the cubic $V = 0$ is the projection of some section of the scroll, so the hyperbola and line are the projection of a section of the scroll, viz. the section through one of the generating lines (there are three such lines) parallel to the basic plane. But it is better to construct 'the cubic' by a different method (using only the basic cubic $U = 0$) which results more immediately from the geometrical theory. Taking the basic plane as the plane of the figure, let O be the node, or foot of the nodal directrix, K the foot of the single directrix, Kk the projection of the single directrix.



k being the projection of the point in which the single directrix meets the plane of the section. Drawing through O any radius vector meeting the basic cubic in P, and the line Kk in r, and producing it to a properly determined point Q, then $OPrQ$ will be the projection of the generating line which meets the nodal directrix, the basic cubic, the single directrix, and the section in the points the projections whereof are O, P, r, Q respectively: and the consideration of the solid figure shows easily that the condition for the determination of the point Q is

$$PQ = Kk \cdot \frac{Pr}{rK}.$$

Hence, starting from the basic cubic and the line Kk, we have a construction for the point Q the locus whereof is the cubic, the projection of a section of the scroll; for the projections of the parallel sections, we have only to vary the length Kk. By what precedes, the construction gives for the locus of Q a cubic having a node at O, and having its asymptotes parallel to those of the basic cubic. As P moves up to K, the distances Pr, rK become indefinitely small; but their ratio is finite, hence the cubic, the locus of Q, does not pass through the point K. The construction shows, however, that it does pass through the points L, M, which are the other two intersections of Kk with the basic cubic; these points L, M are in fact the feet of the generators parallel to the nodal directrix.

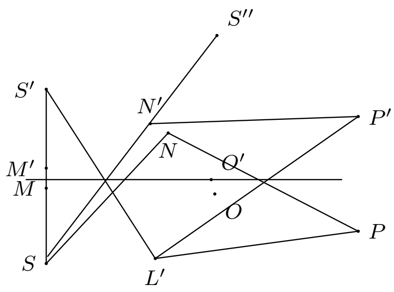
The general conclusion is, that a series of cubics having each of them at one and the same given point a node—having their asymptotes parallel—and besides passing through the same two given points—may be considered as the projections of a series of parallel sections of a cubic scroll; and such a series of cubics will thus afford a delineation of the scroll.

2, Stone Buildings, W.C., April 15, 1863.

329. Note on the Problem of Pedal Curves.

[From the *Philosophical Magazine*, vol. xxvi. (1863), pp. 20, 21.]

It is not, so far as I am aware, generally known that the problem of pedal curves (Steiner's *Fusspuncten-Curve*) was considered by Maclaurin in the *Geometria Organica*, 1720. He appears to have been led to it through an idea such as Sir W. R. Hamilton's Hodograph, or at any rate with a view to a dynamical application, for he remarks, p. 95, "Cum vero geometria quæ curvas ad datum centrum relatas contemplatur in philosophia naturali ad motus corporum et vires evolvendas facilius applicari possit, ... hac sectione considerabimus curvas tanquam ad punctum quodvis datum relatas ex quo ad omnia circumferentiæ puncta radii undique educuntur, et simul perpendicula in illorum punctorum tangentes demittuntur, et rationem radii ad perpendiculum tanquam curvæ characterem usurpabimus." And accordingly, Props. IX. to XII., he considers the problem: Given a point S in the plane of a given curve, to find the locus of the intersection of a tangent of the curve by the perpendicular let fall upon it from the point S; with some special cases, and deductions from it. In particular if the given curve be a circle, the locus in question (or pedal curve) is a curve of the fourth order having a double point S; viz. if S be inside the circle, this is a conjugate or isolated point; but if outside, a double point with two real branches: if S be on the circle, then instead of the double point we have a cusp: it is shown that in each case the pedal curve is in fact an epicycloid. If the given curve be a parabola, then the locus or pedal curve is a curve of the third order, viz. a defective hyperbola having a double point at S, and with its single asymptote perpendicular to the axis of the parabola: some particular cases are specially noticed. If the curve be an ellipse or hyperbola, then, as in the case of the circle, the locus or pedal curve is a curve of the fourth order having a double point at S. And it is moreover shown, Prop. XII., that for any given curve whatever the locus or pedal curve is, in a generalized sense of the term, an epicycloid. This is in fact seen very easily by a mere inspection of the figure. Imagine the curve $O'P'$, rigidly connected with and carrying along with it the point S', to roll on the similar and equal fixed curve OP symmetrically situate on the other side of the common tangent OM or $O'M$; then when P' coincides with P, the point S' is brought to S'', where $SNN'S''$ is the perpendicular from S on the tangent PN or PN', and $SN = N'S''$, that is, $SS'' = 2\,SN$;



and the curve generated by S'' (that is S'), or say the epicycloid the locus of S', is a curve similar to and similarly situate with the pedal curve the locus of N, but of twice the linear magnitude of the pedal curve. Or, what is the same thing, if instead of the given curve we consider a similar and similarly situated curve of twice the linear magnitude (the point S being the centre of similitude), then the epicycloid the locus of S' is the pedal curve of the substituted curve in relation to the point S. It may be added that, in accordance with a theorem of Dandelin's, if rays proceeding from the point S are reflected at the given curve, then the epicycloid (or pedal) in question is the secondary caustic, or an orthogonal trajectory of the reflected rays.

2, Stone Buildings, W.C., June 3, 1863.

330. On Differential Equations and Umbilici.

[From the *Philosophical Magazine*, vol. xxvi. (1863), pp. 373–379 and 441–452.]

I.

Consider the integral equation

$$Az^2 + 2Bz + C = 0,$$

where z is the constant of integration: the derived equation is

$$\begin{aligned}\Omega &= (AC' + A'C - 2BB')^2 - 4(AC - B^2)(A'C' - B'^2)\\ &= (CA' - C'A)^2 - 4(AB' - A'B)(BC' - B'C) = 0;\end{aligned}$$

and if for greater simplicity we write $A = 1$, then the derived equation is

$$\Omega = C'^2 - 4BC'B' + 4CB'^2 = 0,$$

corresponding to the integral equation

$$z^2 + 2Bz + C = 0.$$

Writing the integral equation under the form

$$(z + X)(z + Y) = 0,$$

we have

$$2B = X + Y, \quad C = XY,$$

$$2B' = X' + Y', \quad C' = XY' + X'Y,$$

and the derived equation becomes

$$\Omega = -(X - Y)^2 X'Y'.$$

Hence if we represent the roots X, Y in the form $P \pm Q\sqrt{\square}$, so that $P = -B$, $Q\sqrt{\square} = \sqrt{B^2 - AC}$, Q being the greatest square factor of $B^2 - AC$, then

$$(X - Y)^2 = 4Q^2\square, \quad X',\ Y' = P' \pm \left(Q'\sqrt{\square} + \frac{Q\square'}{2\sqrt{\square}}\right),$$

$$X'Y' = P'^2 - \frac{1}{4\square}\,(2Q'\square + Q\square')^2;$$

and the derived equation is

$$\Omega = -Q^2\big\{4\square P'^2 - (2Q'\square + Q\square')^2\big\} = 0.$$

If B, C, &c. are functions of the coordinates $(x,\ y)$, the equation $z^2 + 2Bz + C = 0$ (z an arbitrary constant) represents a series of curves in the plane of xy; but if we consider z as a coordinate, then the equation represents a surface, and the curves in question are the orthogonal projections on the plane of xy of the sections of the surface by the planes parallel to the plane of xy. To fix the ideas, the plane of xy may be taken to be horizontal, and the ordinates z vertical.

Writing the equation in the form

$$(z + B)^2 - (B^2 - C) = 0,$$

we see that the surface contains upon it the curve $z + B = 0$, $B^2 - C = 0$, which is the line of contact with the circumscribed vertical cylinder: such curve may be termed the envelope, or, when this is necessary, the complete envelope. The equation of the surface has however been taken to be $(z - P)^2 - Q^2\square = 0$ (viz. it has been assumed that $B = -P$, $B^2 - C = Q^2\square$); the envelope thus breaks up into the curve $z - P = 0$, $Q = 0$, taken twice, and the curve $z - P = 0$, $\square = 0$; the former of these is in general a nodal curve on the surface, and it may be spoken of as the nodal curve; the latter of them is the reduced or proper envelope, or simply the envelope. And the terms nodal curve and envelope may also be applied to the curves $Q = 0$ and $\square = 0$, which are the projections on the plane of xy of the first-mentioned two curves respectively. There is however a case of higher singularity which it is proper to consider: suppose that Q and $\square$ have a common factor K, say $Q = KR$, $\square = K\nabla$; the complete envelope $Q^2\square = K^3R^2\nabla = 0$ here breaks up into the nodal curve $R = 0$ twice, the cuspidal curve $K = 0$ three times, and the reduced or proper envelope $\nabla = 0$ once.

Reverting for a moment to the form $(z + X)(z + Y) = 0$, the derived equation $\Omega = -(X - Y)^2X'Y' = 0$ is satisfied by $(X - Y)^2 = 0$; this equation, or say the equation of the envelope, being in fact the singular solution of the differential equation. This assumes however that the differential equation is given in the form in which it is immediately obtained by derivation from the integral equation, without the rejection of factors which are functions of the coordinates $(x,\ y)$ only; it is proper to consider the reduced equation obtained by rejecting such factors. Thus if X and Y are rational functions, the reduced form is $X'Y' = 0$, which is no longer satisfied by the equation

$(X-Y)^2 = 0$. In the before-mentioned case where the roots are $P \pm Q\sqrt{\Box}$ (or $(X-Y)^2 = 4Q^2\Box$), P, Q, and $\Box$ being rational functions of (x, y), the derived equation

$$\Omega = -Q^2\{4\Box P'^2 - (2Q'\Box + Q\Box')^2\} = 0$$

divides out by the factor Q^2, but it does not divide out by $\Box$; the reduced form is therefore

$$4\Box P'^2 - (2Q'\Box + Q\Box')^2 = 0,$$

which is not satisfied by $Q = 0$, while it is still satisfied by $\Box = 0$ (since this gives also $\Box' = 0$); that is, the nodal curve $Q = 0$ is not a solution of the differential equation, but we still have the singular solution $\Box = 0$, which corresponds to the reduced or proper envelope. In the case $Q = KR$, $\Box = K\nabla$ of a cuspidal curve, the above form of the differential equation becomes

$$4K\nabla P'^2 - \{3KK'R\nabla + K^2(2\nabla R' + \nabla' R)\}^2 = 0,$$

which divides out by K; and, when reduced by the rejection of this factor, it is no longer satisfied by the equation $K = 0$, which belongs to the cuspidal curve; that is, neither the nodal curve $R = 0$ nor the cuspidal curve $K = 0$ is a solution of the differential equation, but we still have the singular solution $\nabla = 0$, which corresponds to the reduced or proper envelope. It would appear that the conclusion may be extended to singularities of a higher nature, viz. the factor corresponding to any singular curve which presents itself as part of the complete envelope divides out from the derived equation; and such singular curve does not constitute a solution of the reduced equation, but we have a singular solution corresponding to the reduced or proper envelope.

II.

Consider the differential equation

$$y(p^2-1) + 2mxp = 0,$$

where, to fix the ideas, $m \geq 1$; the integral equation may be taken to be

$$z = (mx + \sqrt{m^2x^2+y^2})(mx^2 + y^2 + x\sqrt{m^2x^2+y^2})^{m-1};$$

or rather, writing for shortness $\Box = m^2x^2 + y^2$, and putting

$$z = (mx + \sqrt{\Box})(mx^2 + y^2 + x\sqrt{\Box})^{m-1} = P + Q\sqrt{\Box},$$

the integral equation is

$$(z-P)^2 - Q^2\Box = 0, \quad \text{or} \quad z^2 - 2Pz + P^2 - Q^2\Box = 0,$$

where

$$P^2 - Q^2\Box = (m^2x^2 - \Box)\{(mx^2+y^2)^2 - x^2\Box\}^{m-1} = -y^{2m}\{y^2 + (2m-1)x^2\}^{m-1}.$$

In the particular case $m = 1$ the equation is

$$z = x + \sqrt{x^2+y^2}, \quad \text{or} \quad z^2 - 2zx - y^2 = 0.$$

Before going further, I remark that, m being a positive integer greater than unity, we have

$$z = P + Q\sqrt{\square} = mx(mx^2+y^2)^{m-1} + \{mx^2+y^2+(m-1)mx^2\}(mx^2+y^2)^{m-2}\sqrt{\square} + \text{\&c.},$$

the subsequent terms being divisible, the rational ones by $\square$, and the irrational ones by $\square\sqrt{\square}$. Hence, observing that

$$mx^2+y^2+(m-1)mx^2 = m^2x^2+y^2 = \square,$$

we see that Q contains the factor $\square$, and the equation $\square = 0$ belongs to a cuspidal curve on the surface. If however $m = 1$, then the equation is $z = x + \sqrt{\square}$, so that $Q = 1$ does not contain the factor $\square$; and $\square = x^2+y^2 = 0$ is not a singular curve on the surface, but is in fact the reduced or proper envelope.

The curve represented by the integral equation will pass through the origin ($x = 0$, $y = 0$) for the value $z = 0$ of the constant of integration. In fact, for this value, the integral equation becomes

$$-y^{2m}\{y^2+(2m-1)x^2\}^{m-1} = 0,$$

which belongs to a set of $2m + (m-1) + (m-1)$ lines coinciding with the lines $y = 0$, $y = ix\sqrt{2m-1}$, and $y = -ix\sqrt{2m-1}$ respectively. The directions at the origin are therefore $p = 0$, $p = \pm i\sqrt{2m-1}$, which are the same values of p as are obtained from the differential equation; viz. since this is satisfied identically at the point in question, proceeding to the derived equation, we have

$$p(p-1) + 2mp = 0,$$

that is

$$p(p^2+2m-1) = 0;$$

but it is to be observed that these values of p are different from the values given by the equation $\square = m^2x^2+y^2 = 0$, which are $p = \pm im$. The reason is that the curve $\square = 0$ being, as was shown, a cuspidal curve on the surface, the equation $\square = 0$ is not a solution of the differential equation.

If however $m = 1$, then the integral equation gives at the origin no longer three values of p, but only the value $p = 0$. The differential equation however gives, as in the general case, three values; viz. we have $p(p^2+1) = 0$; and the values $p = \pm i$ obtained from the factor $p^2+1 = 0$ are precisely the values of p obtained from the equation $\square = x^2+y^2 = 0$, which in the case now under consideration belongs to the reduced or proper envelope of the surface, and is therefore the singular solution of the differential equation.

III.

The two curves of curvature which pass through any given point of a surface are distinct curves, not branches of one indecomposable curve. In fact if P, Q are the two curves of curvature for a point A, then for a point A' on P the two curves of curvature will be P, Q'; and if P, Q were branches of an indecomposable curve, then P, Q' would also be branches of an indecomposable curve, and we should have P a branch of two different indecomposable curves, which is of course impossible. In the case of an umbilicus, the two curves P and Q coincide together; or, as we may express it, the curves of curvature through an umbilicus are the duplication of a single, in general indecomposable, curve; and in general this curve has at the umbilicus a trifid node. I use this expression to denote a point at which there are three distinct tangents, or, more accurately, three distinct directions of the curve: an ordinary triple point is of necessity a trifid node, but not conversely. The umbilicus of an ellipsoid or other quadric surface is a peculiar exceptional case.

In support of the foregoing conclusions, consider a surface having an umbilicus at the origin, and take $z = 0$ as the equation of the tangent plane at that point; the equation of the surface in the neighbourhood of the umbilicus will be

$$z = \tfrac{1}{2}k(x^2 + y^2) + \tfrac{1}{6}(ax^3 + 3bx^2y + 3cxy^2 + dy^3);$$

so that, writing as usual p and q for the first, and r, s, t for the second, differential coefficients of z, we have

$$\begin{aligned}
p &= kx + \tfrac{1}{2}(ax^2 + 2bxy + cy^2),\\
q &= ky + \tfrac{1}{2}(bx^2 + 2cxy + dy^2),\\
r &= k + ax + by,\\
s &= bx + cy,\\
t &= k + cx + dy.
\end{aligned}$$

The differential equation of the curves of curvature projected on the plane of xy is

$$\frac{dy^2}{dx^2}\{(1+q^2)s - pqt\} + \frac{dy}{dx}\{(1+q^2)r - (1+p^2)t\} - \{(1+p^2)s - pqr\} = 0;$$

and substituting therein the foregoing values of p, q, r, s, t, but attending only to the terms of the lowest order in (x, y), and using moreover in the sequel p in the place of $\dfrac{dy}{dx}$, the equation becomes

$$(bx + cy)(p^2 - 1) + \{(a-c)x + (b-d)y\}p = 0;$$

which may be taken as the differential equation of the curves of curvature at and in the neighbourhood of the umbilicus. The equation is satisfied identically by the values $x = 0$, $y = 0$, which correspond to the umbilicus; and to find p, we have to differentiate the equation, and then substitute these values of x and y; we thus obtain

$$(b + cp)(p^2 - 1) + \{(a-c) + (b-d)p\}p = 0,$$

or, what is the same thing,

$$p(a + 2bp + cp^2) - (b + 2cp + dp^2) = 0,$$

a cubic equation for the determination of p.

I remark that we may without loss of generality write $d = 0$: but to simplify the investigation, I suppose in the first instance that we have also $b = 0$; this comes to assuming that one of the three planes $ax^3 + 3bx^2y + 3cxy^2 + dy^3 = 0$ bisects the angle formed by the other two planes. The differential equation consequently is

$$cy(p^2 - 1) + (a - c)xp = 0;$$

or, putting for shortness

$$m = \tfrac{1}{2} \cdot \frac{a - c}{c},$$

it is

$$y(p^2 - 1) + 2mxp = 0,$$

which is the differential equation previously considered. Hence, writing now h in the place of z, the equation of the curve of curvature in the neighbourhood of the umbilicus is

$$h = (mx + \sqrt{\Box})(mx^2 + y^2 + x\sqrt{\Box})^{m-1}, \; = P + Q\sqrt{\Box},$$

where $\Box = m^2x^2 + y^2$; or, what is the same thing, the equation is

$$h^2 - 2Ph + P^2 - Q^2\Box = 0;$$

and the equation (in the neighbourhood of the umbilicus) of the curve through the umbilicus is

$$P^2 - Q^2\Box = -y^{2m}\{y^2 + (2m - 1)x^2\}^{m-1} = 0;$$

so that the umbilicus is a trifid node. In the case however of an ellipsoid or other quadric surface, we have $m = 1$, so that the equation of the curve of curvature in the neighbourhood of the umbilicus is

$$h = x + \sqrt{x^2 + y^2},$$

or, what is the same thing,

$$h^2 - 2hx - y^2 = 0 :$$

and for the curve through the umbilicus, in the neighbourhood of the umbilicus, the equation is $y^2 = 0$, so that there is only a single direction of the curve of curvature. The differential equation gives, however, at the umbilicus $p(p^2 + 1) = 0$; the value $p = 0$ is that which corresponds to the curve of curvature; the other two values $p = \pm i$ correspond to the curve (pair of lines) $x^2 + y^2 = 0$, which is the envelope of the curves of curvature, or, more accurately, the envelope of the projections of the curves of curvature on the tangent plane at the umbilicus.

Blackheath, October 17, 1863.

IV.

The differential equation for the curves of curvature in the neighbourhood of an umbilicus was obtained in a form such as

$$(bx + cy)(p^2 - 1) + 2(fx + gy)p = 0;$$

and it was only because this equation did not appear to be readily integrable, that I considered, instead of it, the particular form

$$y(p^2-1)+2mxp=0.$$

But the general equation can be integrated; and the result presents itself in a simple form. For, returning to the differential equation

$$(bx+cy)(p^2-1)+2(fx+gy)p=0,$$

and assuming

$$\frac{bx+cy}{fx+gy}=\frac{-2v}{v^2-1},$$

we have

$$(bx+cy)(p^2-1)+2(fx+gy)p=0$$

gives

$$\frac{p^2-1}{p}=-\frac{2v}{v^2-1}, \text{ or } (p-v)(vp+1)=0,$$

and we may write

$$p-v=0.$$

Assuming also

$$v=ux, \quad \text{or} \quad u=\frac{y}{x},$$

the relation between u and v is

$$\frac{b+cu}{f+gu}=\frac{-2v}{v^2-1},$$

or, as this may be written,

$$v^2+\frac{2(f+gu)}{b+cu}v-1=0,$$

giving

$$v=\frac{-(f+gu)-\sqrt{(b+cu)^2+(f+gu)^2}}{b+cu},$$

where for convenience the radical has been taken with a negative sign. We have moreover

$$p=\frac{b(v^2-1)+2fv}{c(v^2-1)+2gv}.$$

The equation $p-v=0$, substituting for y its value ux, then becomes

$$x\frac{du}{dx}+u-v=0,$$

or, as this may be written,

$$\frac{dx}{x}-\frac{du}{u-v}=0,$$

or, what is the same thing,

$$\frac{dx}{x} + \frac{dv - du}{v - u} - \frac{dv}{v - u} = 0.$$

But

$$v - u = v - p = v - \frac{b(v^2 - 1) + 2fv}{c(v^2 - 1) + 2gv} = \frac{V}{c(v^2 - 1) + 2gv},$$

where

$$V = v\{c(v^2 - 1) + 2gv\} - \{b(v^2 - 1) + 2fv\} = (b + cv)(v^2 - 1) + 2(f + gv)\,v,$$

(with sign as above; the displayed identity is

$$V = (b + cv)(v^2 - 1) + 2(f + gv)v).$$

and the differential equation takes thus the form

$$\frac{dx}{x} + \frac{dv - du}{v - u} - \frac{\{c(v^2 - 1) + 2gv\}\,dv}{V} = 0,$$

and hence, writing

$$V = (b + cv)(v^2 - 1) + 2(f + gv)v = c(v - \alpha)(v - \beta)(v - \gamma),$$

and

$$\frac{c(v^2 - 1) + 2gv}{V} = \frac{c(v^2 - 1) + 2gv}{c(v - \alpha)(v - \beta)(v - \gamma)} = \frac{A}{v - \alpha} + \frac{B}{v - \beta} + \frac{C}{v - \gamma},$$

so that

$$A = \frac{c(\alpha^2 - 1) + 2g\alpha}{c(\alpha^2 - 1) + 2g\alpha + 2\{f + (b + g)\alpha + c\alpha^2\}},$$

with the like values for B and C—values which are such that $A + B + C = 1$,—the integral equation is

$$\text{const.} = x(v - u)(v - \alpha)^{-A}(v - \beta)^{-B}(v - \gamma)^{-C},$$

or, substituting for $v - u$ its value,

$$= x \cdot \frac{V}{c(v^2 - 1) + 2gv}(v - \alpha)^{-A}(v - \beta)^{-B}(v - \gamma)^{-C}.$$

But

$$\text{const.} = x\,\{c(v^2 - 1) + 2gv\}^{-1}(v - \alpha)^{1-A}(v - \beta)^{1-B}(v - \gamma)^{1-C}.$$

Substituting for v its value, if for shortness $U = (b + cu)^2 + (f + gu)^2$, and hence $\sqrt{U}$ = the radical $\sqrt{(b + cu)^2 + (f + gu)^2}$, then

$$v^2 = \frac{2(f + gu)^2 + (b + cu)^2 + 2(f + gu)\sqrt{U}}{(b + cu)^2},$$

and

$$c(v^2 - 1) + 2gv = \frac{c\{(f + gu)^2 + (f + gu)\sqrt{U}\} + g\{-(f + gu) - \sqrt{U}\}(b + cu)}{(b + cu)^2}$$

(intermediate expressions; the reduction reads as in the original),

$$v - \alpha = \frac{-(f + gu) - \sqrt{U} - \alpha(b + cu)}{b + cu}, \quad \&\text{c.}$$

Substituting these values, and observing that the exponent of $b + cu$ is

$$(-2 + 1 - A + 1 - B + 1 - C, \; = 1 - A - B - C) = 0,$$

the integral equation is

$$\text{const.} = x\,(f + gu + \sqrt{U})^{-1} \times$$
$$(f + gu + \alpha(b + cu) + \sqrt{U})^{1-A}(f + gu + \beta(b + cu) + \sqrt{U})^{1-B}(f + gu + \gamma(b + cu) + \sqrt{U})^{1-C},$$

or, observing that the exponent 1 of x is

$$= -1 + (1 - A) + (1 - B) + (1 - C),$$

and putting for shortness $\square = (fx + gy)^2 + (bx + cy)^2$, the integral equation finally is

$$\text{const.} = (fx + gy + \sqrt{\square})^{-1} \times$$
$$(fx+gy+\alpha(bx+cy)+\sqrt{\square})^{1-A}(fx+gy+\beta(bx+cy)+\sqrt{\square})^{1-B}(fx+gy+\gamma(bx+cy)+\sqrt{\square})^{1-C},$$

where the quantities α, β, γ, A, B, C are given by

$$(b + cv)(v^2 - 1) + 2(f + gv) = c(v - \alpha)(v - \beta)(v - \gamma),$$

$$\frac{c(v^2 - 1) + 2gv}{c(v - \alpha)(v - \beta)(v - \gamma)} = \frac{A}{v - \alpha} + \frac{B}{v - \beta} + \frac{C}{v - \gamma}.$$

Consider the curve

$$0 = (fx+gy+\alpha(bx+cy)+\sqrt{\square})^{1-A}(fx+gy+\beta(bx+cy)+\sqrt{\square})^{1-B}(fx+gy+\gamma(bx+cy)+\sqrt{\square})^{1-C},$$

which corresponds to the value $= 0$ of the constant. If, for instance, this equation gives

$$fx + gy + \alpha(bx + cy) + \sqrt{\square} = 0,$$

or say

$$(bx + cy)\{(bx + cy)(\alpha^2 - 1) + 2(fx + gy)\alpha\} = 0;$$
$$(bx + cy)(\alpha^2 - 1) + 2(fx + gy)\alpha = 0.$$

But we have

$$(b + c\alpha)(\alpha^2 - 1) + 2(f + g\alpha)\alpha = 0,$$

and the equation therefore is

$$(bx + cy)(f + g\alpha) - (fx + gy)(b + c\alpha) = 0;$$

that is

$$(cf - bg)(y - \alpha x) = 0;$$

or simply $y - \alpha x = 0$; that is, the directions of the curve at the origin, or point $x = 0$, $y = 0$, are given by the equations $y - \alpha x = 0$, $y - \beta x = 0$, $y - \gamma x = 0$. This is right, since from the differential equation we obtain at the origin

$$(b + cp)(p^2 - 1) + 2(f + gp)p, \; = c(p - \alpha)(p - \beta)(p - \gamma), \; = 0.$$

V.

The particular case of the equation

$$y(p^2-1)+2mxp=0$$

is obtained from the general equation by writing therein $b=0$, $c=1$, $g=0$, $f=m$; we have therefore

$$v(v^2+2m-1)=(v-\alpha)(v-\beta)(v-\gamma),$$

or say

$$\alpha=0,\quad \beta=i\sqrt{2m-1},\quad \gamma=-i\sqrt{2m-1};$$

$$\frac{v^2-1}{v(v^2+2m-1)}=\frac{A}{v}+\frac{B}{v-i\sqrt{2m-1}}+\frac{C}{v+i\sqrt{2m-1}},$$

giving

$$A=-\frac{1}{2m-1},\quad B=C=\frac{m}{2m-1}.$$

The integral equation thus is

$$\text{const.}=(mx-\sqrt{\square})^{-1}(mx+\sqrt{\square})^{\frac{2m}{2m-1}}\,\{(mx+i\sqrt{2m-1}\,y+\sqrt{\square})(mx-i\sqrt{2m-1}\,y+\sqrt{\square})\}^{\frac{m-1}{2m-1}},$$

where $\square=m^2x^2+y^2$; or, observing that

$$\begin{aligned}(mx+i\sqrt{2m-1}\,y+\sqrt{\square})(mx-i\sqrt{2m-1}\,y+\sqrt{\square})&=(mx+\sqrt{\square})^2+(2m-1)\,y^2\\&=2m(mx^2+y^2+x\sqrt{\square}),\end{aligned}$$

the integral equation is

$$\text{const.}=(mx+\sqrt{\square})^{\frac{1}{2m-1}}(mx^2+y^2+x\sqrt{\square})^{\frac{m-1}{2m-1}},$$

or, what is the same thing,

$$\text{const.}=(mx+\sqrt{\square})(mx^2+y^2+x\sqrt{\square})^{m-1},$$

the result given in the former part of the present paper.

VI.

I annex the following *à posteriori* verification of the solution

$$\text{const.}=(mx+\sqrt{\square})(mx^2+y^2+x\sqrt{\square})^{m-1}$$

of the particular equation

$$y(p^2-1)+2mxp=0.$$

Putting for shortness

$$A = mx + \sqrt{\square}, \quad B = mx^2 + y^2 + x\sqrt{\square},$$

where it will be remembered that

$$\square = m^2x^2 + y^2, \quad 2mB = A^2 + (2m-1)\,y^2.$$

The integral equation may be written

$$h = P + Q\sqrt{\square} = U = AB^{m-1};$$

and we have

$$\frac{U'}{U} = \frac{A'}{A} + (m-1)\frac{B'}{B} = \frac{\Theta}{AB},$$

if

$$\Theta = \frac{A'}{A}AB + (m-1)\frac{B'}{B}AB,$$

$$= \frac{1}{AB}\{A' \cdot AB + (m-1)B' \cdot AB \cdot \tfrac{1}{B} \cdot A\}.$$

But we have

$$A'\sqrt{\square} = m\sqrt{\square} + m^2x + yp = mA + yp,$$

and

$$\begin{aligned} B'\sqrt{\square} &= (2mx + 2yp)\sqrt{\square} + \square + x(m^2x + yp) \\ &= 2m^2x^2 + y^2 + xyp + (2mx + 2yp)\sqrt{\square} \\ &= A^2 + pyx + 2\sqrt{\square} \cdot (\text{terms as in original}), \end{aligned}$$

and the value of Θ thus is

$$\Theta = \frac{1}{\sqrt{\square}}\left\{A^2B \cdot \frac{1}{A} + (m-1) \cdot \frac{A^2 + py(x + 2\sqrt{\square})}{B} \cdot A\right\}.$$

$$= \frac{1}{2m\sqrt{\square}}\Big\{[A^2 + (2m-1)\,y^2]\,(mA + yp) + (A^2 - 2m)\,[A^2 + py(x + 2\sqrt{\square})]\Big\},$$

where the expression in { } is

$$= (2m^2 - m)\,A\,(A^2 + y^2) + yp\{A^2 + (2m-1)y^2 + (2m^2 - 2m)\,A(x + 2\sqrt{\square})\}.$$

Here the coefficient of yp is $= (2m^2 - m)(A^2 + y^2)$; in fact we have identically

$$A^2 + y^2 - 2A\sqrt{\square} = 0,$$

and thence

$$(2m^2 - 3m + 1)(A^2 + y^2) - 2(2m-1)(m-1)\,A\sqrt{\square} = 0,$$

that is

$$(2m^2 - m - 1)\,A^2 + (2m^2 - 3m + 1)\,y^2 - (2m-2)\,A\{A^2 + (2m-1)\sqrt{\square}\} = 0;$$

[Continues on p. 126; this transcript ends mid-derivation at p. 125.]